

RUIJIE NIU

Pittsburgh, PA | (412) 808-0916 | run30@pitt.edu

EDUCATION

University of Pittsburgh, School of Public Health

Expected May 2027

M.S. in Biostatistics | Cumulative GPA: **3.875/4.000** | Pittsburgh, PA

Honors: Pitt Public Health Futures Award; Biostatistics Achievement Award.

Selected completed coursework: Mathematical Methods for Statistics (A+); Applied Regression Analysis (A+); Advanced R Computing (A+); Machine Learning (A).

In progress (Fall 2026): Causal Inference in Educational Research; Longitudinal and Clustered Data Analysis; Applied Survival Analysis: Methods and Practice.

Xi'an Jiaotong-Liverpool University (XJTLU)

Jun 2025

B.S. in Biomedical Statistics, Honours | Dual degrees: XJTLU and University of Liverpool.

Award: Scholarship for Outstanding Class Cadres.

RESEARCH INTERESTS

Causal mediation and negative controls; difference-in-differences; statistical methods for clinical studies; Bayesian modeling of high-dimensional biomedical data.

SELECTED RESEARCH EXPERIENCE

Causal Mediation with Negative Controls and DID

Jan 2026 - Present

Student Research Assistant | University of Pittsburgh, Department of Biostatistics | Supervisor: Qiong Wu, PhD

- Take primary responsibility for identification derivations, estimator implementation, and simulation design under faculty supervision; study total, controlled direct, and natural direct/indirect effects under stated negative-control assumptions.
- Implemented individual-level plug-in/g-computation estimators and participant bootstrap with full nuisance-model refitting; checked transformed-product/four-term equivalence for binary and linear-mean continuous mediators.
- Completed 27,000 Monte Carlo replications across 27 scenarios, with 1,000 bootstrap resamples per replication; assessed bias, coverage, and standard-error calibration under confounding, interactions, and model misspecification.
- Distinguished precision loss from noisy valid controls versus bias from loading mismatch; examined direct natural-effect estimation when total-effect subtraction fails in prespecified simulation profiles.

Bayesian Factor Modeling for Single-Cell Perturbation Data

Jun 2026 - Present

Graduate Research Collaborator | University of Pittsburgh, Department of Medicine | Mentor: Qiyiwen Zhang, PhD

- Reproduced core guided sparse factor analysis (GSFA) workflows using the authors' code for LUHMES (8,708 cells) and CD8+ T-cell (24,955 cells) data, with 6,000 genes per dataset.
- Completed 180 GSFA fits across 90 paired simulation cases; compared Welch/edgeR-QLF baselines and independent LUHMES MAST/clustering analyses, documenting false-discovery behavior and replication limits to inform planned extensions.

Clinical Biostatistics: Oncology and Pediatric Endocrinology

Nov 2025 - Present

Biostatistics Research Assistant | University of Pittsburgh / UPMC

Oncology | Mentor: Hong Wang, PhD

- Construct oncology cohorts and response/time-to-event endpoints; apply Kaplan-Meier estimation, log-rank tests, exact binomial confidence intervals, and follow-up summaries, delivering statistical results and reports to clinical investigators.

Pediatric endocrinology | Mentor: Qiong Wu, PhD

- Analyze sex differences in cardiometabolic risk four years after youth-onset type 1 diabetes diagnosis; define cohorts/endpoints, assess attrition and missingness, fit logistic regression and interactions, and prepare adjusted-effect tables and figures.

ADDITIONAL RESEARCH EXPERIENCE

Foundation Models in Emergency and Critical Care

Apr 2026 - Present

Research Contributor / Co-author | University of Minnesota, Computational Health Sciences | Mentor: Feng Xie, PhD | Remote

- Synthesized 49 eligible studies under PRISMA-ScR/JBI guidance across nine evidence dimensions, focusing on clinical utility, decision-making, and deployment gaps; co-authored a scoping-review manuscript submitted to Critical Care.

Diabetes Nutrition and HbA1c Modeling

Mar 2024 - Sep 2024

Lead Analyst | Xi'an Jiaotong-Liverpool University | Advisor: Hui Zhang, PhD

- Built an R/SAS workflow for cohort cleaning, nutrition-HbA1c modeling, demographic comparisons, and sensitivity analyses; prepared tables and figures for a first- and corresponding-author conference-proceedings publication.

Undergraduate Thesis: Layered Tuberculosis Agent-Based Model

May 2024

Xi'an Jiaotong-Liverpool University | Advisor: Tenglong Li, PhD

- Built a Python/Numba agent-based model with home, school, work, and community contact layers; calibrated to 12 months of 2023 notifications in two countries and assessed scenario uncertainty across random seeds.

Healthcare Time-Series Deep Learning

Jan 2024 - Apr 2024

Kaggle HMS Competition Project | Python, PyTorch

- Reproduced ChronoNet, WaveNet, EfficientNet, and Transformer approaches for noisy EEG classification in a shared Python/PyTorch benchmark pipeline with external-data augmentation.

PUBLICATIONS AND SUBMITTED MANUSCRIPT

Published conference-proceedings articles

Niu, R., & Xia, J. (2025). Vitamin and Mineral Combined in Diabetes: Modulating HbA1c Across Different Demographics. *Applied and Computational Engineering*, 132, 257-268. doi:10.54254/2755-2721/2024.20798. (First and corresponding author.)

Niu, R. (2024). The role of plasma trace elements in hypertension among elderly populations in China: A cross-sectional analysis using CLHLS data. *Theoretical and Natural Science*, 70, 18-23. doi:10.54254/2753-8818/2024.18203. (Sole and corresponding author.)

Submitted manuscript

Co-author. Mapping the Clinical Utility of Foundation Models in Emergency and Critical Care: A Scoping Review of Decision-Making, Implementation, and Research Gaps. Submitted to *Critical Care*, 2026.

CLINICAL AND INDUSTRY EXPERIENCE

Tasly Pharma | Biostatistics Intern / SAS Programmer

Jun 2024 - Sep 2024

- Supported data cleaning, SAS programming, derived-variable checks, and quality-control listings across four Phase II/III studies; reviewed eCRF/SAP specifications and draft statistical tables, listings, figures, and reports under statistician supervision.

Tianjin Fourth Center Hospital | Medical Records / Information Intern

Jul 2023 - Aug 2023

- Resolved patient medical-record access issues through verification and coordination; reconciled data for departmental reporting and validated medication fields for a pharmacy-linked QR prescription-information service.

TECHNICAL SKILLS

Programming: R (tidyverse, survival, survminer, ggplot2, R Markdown); SAS; Python (pandas, NumPy, Numba, PyTorch); LaTeX.

Methods: Causal mediation; negative controls/DID; plug-in/g-computation; participant bootstrap; Monte Carlo evaluation; survival analysis; generalized linear models; Bayesian factor models.

Research workflows: Reproducible statistical reporting; Slurm-based computation; clinical dataset construction and quality control; clinician-facing interpretation.

Languages: Chinese (native), English.